

Burak Şen

Munich, Germany | github.com/burakssen | linkedin.com/in/burak-ssen | burakssen.com

TECHNICAL SKILLS

- **Programming Languages:** Python, TypeScript, JavaScript, Zig
- **Web Technologies:** React.js, HTMX, Node.js, Express.js, Bun
- **Databases:** PostgreSQL, SQLite
- **AI & Machine Learning:** TensorFlow, PyTorch, ONNX, Jax
- **DevOps & Tools:** Postman, Docker, Git, Github Actions, Gitlab, Gitlab CI/CD

EXPERIENCE

TUM CommonRoad | Full Stack Developer

Jul 2024 — Present

Django | React | HTMX | PostgreSQL | Minio | Gitlab CI/CD

Munich, Germany

- Designed and maintained a web framework serving **500+ active users** with **99% uptime**.
- Centralized application utilities by creating the **commonx** base package.
- Streamlined page creation workflows through efficient Markdown rendering.

Zant-Foundation | Volunteer Software Developer

Apr 2025 — Present

Zig | ONNX

Italy, (Remote)

- Contributed to an open-source SDK for optimized Neural Network deployment on edge devices.
- Developed a **Zig-based backend** and applied code generation from micro-instructions.
- Achieved a **30% reduction in memory usage** via static allocation planning for embedded systems.

TUM CIT | Assistant Student (Software Development)

Dec 2022 — Oct 2023

React.js | Express.js | CouchDB | Docker | Redis | Nginx | Shibboleth SP

Munich, Germany

- Architected and deployed a module creator application for the TUM School of CIT.
- Enhanced workflow efficiency by **10%** for over 100 faculty members.

Map E-Commerce & Data Services Inc. | Software Developer

Apr 2022 — Aug 2022

Node.js | C++ | PHP | Go | MySQL | XML | EDI | JSON | VDA

Istanbul, Turkey

- Engineered a high-throughput system processing **150 million EDI messages** annually.
- Improved data interoperability and reduced system processing time by **15%**.
- Maintained a **Go backend** to ensure scalability for hundreds of products.

IBM | Internship in Software Development

Jul 2021 — Jan 2022

R | IBM BPM | ASPX | .NET | MS-SQL | Java | JavaScript

Istanbul, Turkey

- Analyzed **10M+ row datasets** in R to optimize national-level HR policies.
- Enhanced a BPM platform for 50+ clients, maintaining **99.9% uptime** and adding bilingual support.
- Expanded automotive finance solutions and POS integrations using **ASP.NET and MS-SQL**.

PROJECTS

6soz

May 2026 — Present

Zig | Raylib

(github.com/burakssen/6soz)

- Developed a retro gaming emulator in **Zig** with an initial implementation of the **Nintendo Entertainment System (NES)** architecture and cartridge loading pipeline.
- Implemented core emulation subsystems including **CPU instruction execution, memory mapping, ROM parsing, and frame rendering** for the NES platform.
- Designed the emulator with a focus on **performance, low-level systems programming, and cross-platform portability** using Zig's manual memory management and compile-time capabilities.

pkl-zig

Aug 2025 — Present

Zig | Pkl

(github.com/burakssen/pkl-zig)

- Developed native **Zig language bindings for Apple's Pkl configuration language**, enabling **highly performant, type-safe programmatic configuration management**.

- Implemented a **non-blocking, asynchronous transport layer** leveraging Zig 0.16.0's updated I/O interfaces to ensure **efficient, concurrent child process communication**.
- Engineered a robust **message decoding system** to map Pkl's dynamic evaluation outputs seamlessly into Zig's strictly typed, custom-allocated data structures.

JAXFluids Lab

Apr 2025 — Nov 2025

React | Python | Jax | Onnx

(dbezgin.github.io/jaxfluids-lab)

- Built an **interactive web platform** combining JAX-Fluids simulations with ONNX inference, enabling **real-time experimentation** in the browser.
- Implemented **parameter tuning and visualization pipelines**, reducing iteration time for simulation experiments.
- Optimized frontend performance for more scalability.

jax2onnx

Apr 2025 — Nov 2025

Python | Jax | Onnx

(github.com/enpasos/jax2onnx)

- Contributed to and open-source tool to convert **JAX, Flax (NNX), and Equinox** models into ONNX format.
- Extended **jax2onnx** operator coverage by implementing 10+ JAX primitives, enabling conversion of **previously unsupported** model architectures.

zff | Zig Flip Fluid Simulation

Jan 2024 — Mar 2024

Zig | Raylib | WebAssembly

(burakssen.com/zff)

- Ported a **FLIP-based fluid simulation to WebAssembly**, enabling **real-time interactive execution** in the **browser without server-side compute**.
- Realized stable **real-time performance** while maintaining **numerical accuracy** in a constrained **WASM environment**.
- Boosted engagement with mouse-input obstacle manipulation.

boids

Jan 2024 — Feb 2024

C++ | Raylib | WebAssembly

(burakssen.com/boids)

- Simulated flocking behavior in C++ using WebAssembly for online access.
- Modeled realistic entity interactions based on separation, alignment, and cohesion rules.

EDUCATION

Technical University of Munich (TUM)

Munich, Germany

Master's of Science in Informatics

Oct 2022 — Present

Istanbul Technical University (ITU)

Istanbul, Turkey

Bachelor's of Science in Computer Engineering

Sep 2017 — Jul 2022